Parshvanath Charitable Trust's
A. P. SHAH INSTITUTE OF TECHNOLOGY, THANE
(All Programs Accredited by NBA)
Department of Information Technology

SmartClaim AI: Intelligent Vehicle Damage Detection & Insurance Claim Estimator

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1. Introduction

SmartClaim AI is an AI-powered system that turns a single photograph of a damaged car, together with a few vehicle and policy details, into an instant, itemised repair estimate and insurance claim settlement.

- Detects **where** the damage is and **how severe** it is
- Finds the **damaged car parts** and the **type of damage**
- Prices each part: parts, labour, paint and GST
- Applies **IRDAI** depreciation and deductible rules
- Shows what the **insurer pays** and what **you pay**

1 photo
input

3 AI models
YOLO

~2 s
per estimate

2. Problem Statement

- **Slow:** the claim amount is unknown until a surveyor inspects the car.
- **Inconsistent:** two surveyors can value the same damage differently.
- **Opaque:** customers rarely understand depreciation and deductibles.
- **Claim data can't price damage:** on a public claims dataset a model predicting repair cost from damage severity scored $R^2 = -0.14$ (worse than guessing the average).

Problem statement: Design a system that takes a photo of a damaged car and basic vehicle and policy details, identifies the damaged parts and the type of damage, and produces an itemised, explainable repair estimate and claim settlement consistent with Indian motor-insurance rules.

3. Objectives

1. To build and compare deep-learning models that classify damage **location** (front/rear) and **severity** (normal, breakage, crushed).
2. To segment **which car part** is damaged and **what type** of damage it has.
3. To attribute each damage region to the car part it lies on.
4. To design a transparent **pricing engine** using part prices, car segment, labour rates, paint and GST.
5. To apply **IRDAI rules**: depreciation, deductibles, zero-dep cover and total loss.
6. To deliver the complete pipeline as an easy-to-use **web application**.

4. Scope

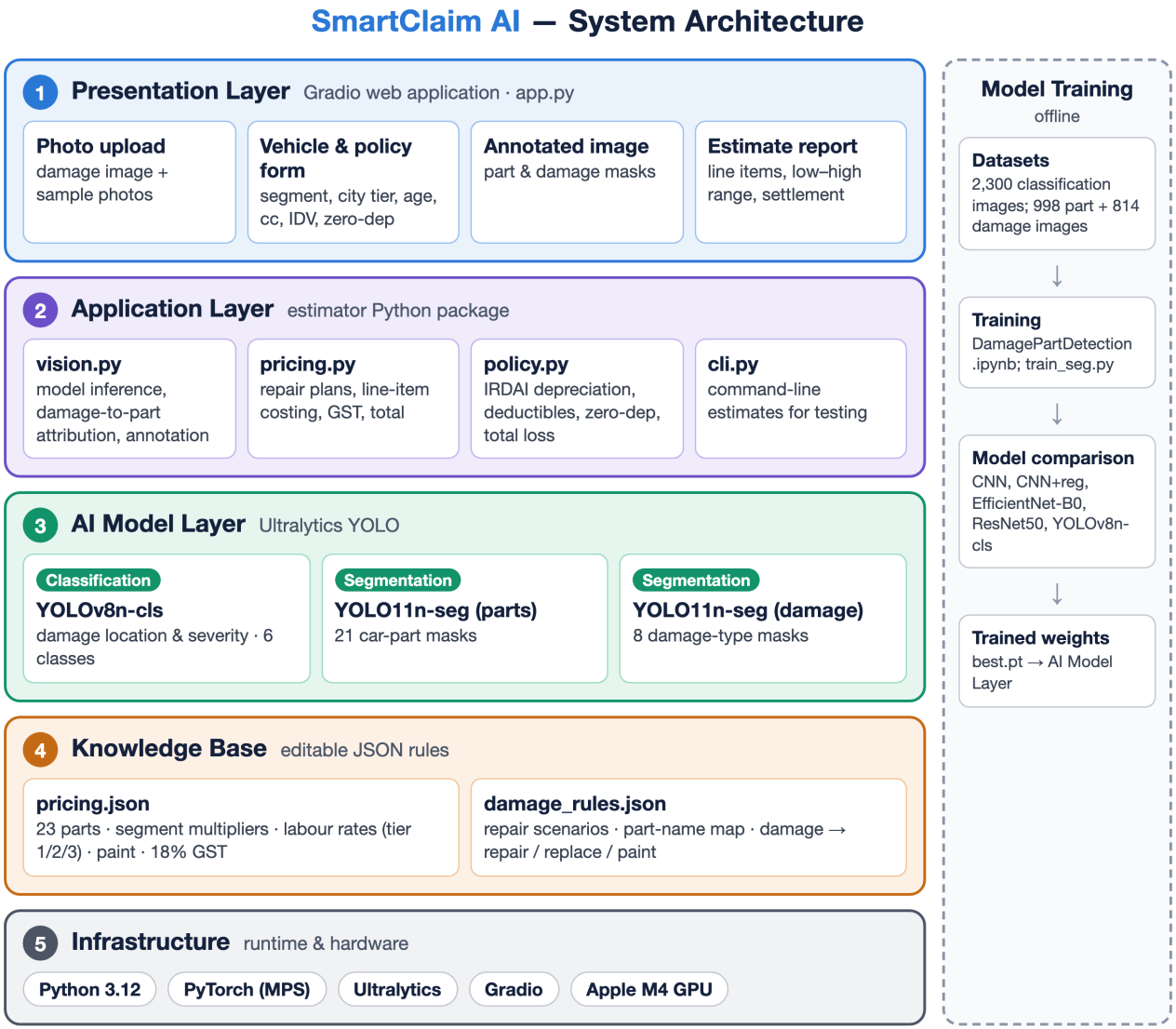
- Can be used to assess damage to **hatchback, sedan, SUV and luxury** cars from one front or rear photo.
- Can be used to classify damage into **6 classes**: front / rear × breakage, crushed, normal.
- Can be used to identify **21 car parts** such as bumpers, bonnet, doors, fenders, lamps and glass.
- Can be used to detect **8 damage types**: dent, scratch, broken part, missing part, crack and more.
- Can be used to generate an itemised repair estimate from **23 priced parts** with city-tier labour rates.
- Can be used to compute the settlement as per **IRDAI** depreciation, deductibles, zero-dep and total loss.
- Can be used through a **web application** to get an instant first estimate.
- Can be extended with multi-photo assessment, live prices, fraud screening and a mobile app.

5. Literature Survey

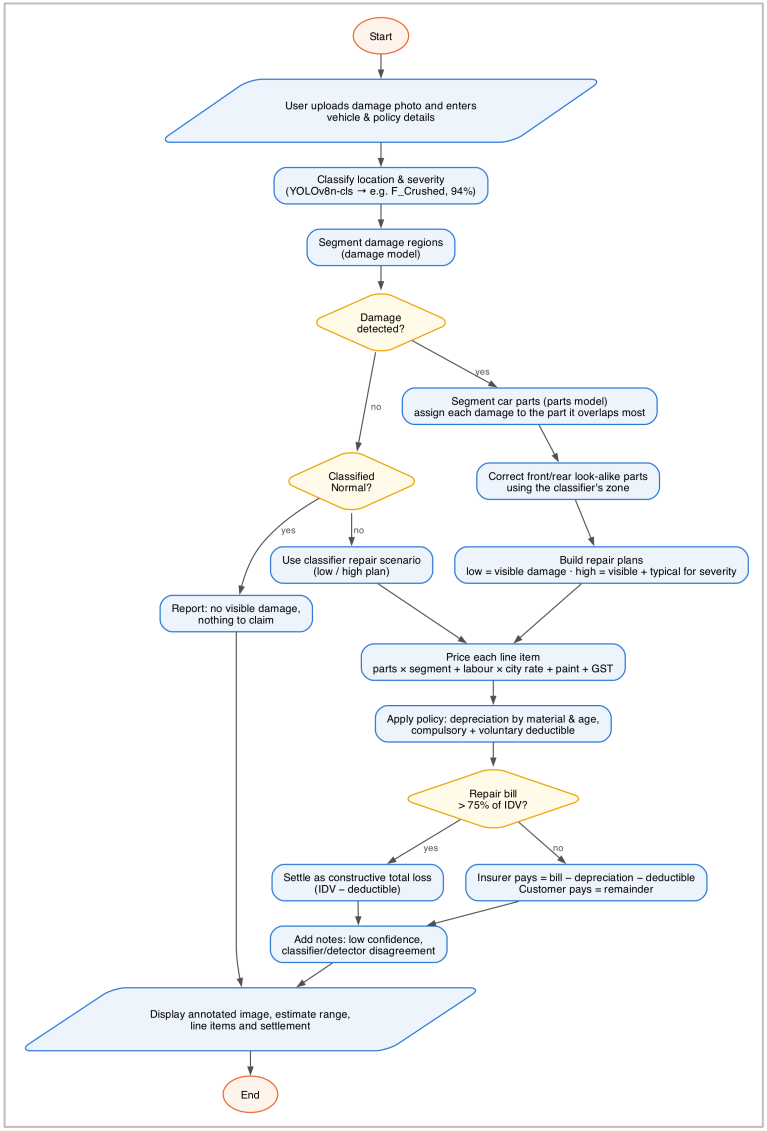
Paper (year)	Key idea	Use in SmartClaim AI
Patil et al., Car damage classification (2017)	CNN + transfer learning on ~1,500 images	Pretrained backbones, augmentation
He et al., ResNet (2016)	Residual connections, deep nets	ResNet50 baseline: 73.9%
Tan & Le, EfficientNet (2019)	Compound model scaling	EfficientNet-B0 baseline: 60.7%
Redmon et al., YOLO (2016)	Single-stage real-time detection	All 3 models are YOLO
He et al., Mask R-CNN (2017)	Per-object pixel masks	Masks for damage-to-part overlap
Wang et al., CarDD (2023)	Car damage segmentation dataset	Damage types and benchmark

Gap: existing work stops at detecting damage; none turns it into a priced, itemised repair plan with Indian insurance rules.

6. Proposed System: Architecture



7. Workflow



1. Upload photo + vehicle & policy details
2. Classify location & severity
3. Segment parts and damage
4. Attribute damage to parts
5. Build low & high repair plans
6. Price: parts, labour, paint, GST
7. Apply depreciation, deductible, total-loss check
8. Show annotated photo & settlement

8. Models & Algorithms

Stage	Model / Algorithm	Output
Location & severity	**YOLOv8n-cls** (CNN, transfer learning, softmax)	6 classes + confidence
Car parts	**YOLO11n-seg** instance segmentation	21 part masks
Damage types	**YOLO11n-seg** instance segmentation	8 damage masks
Attribution	Mask overlap $\|D \cap P\| / \|D\| \geq 0.2$	Damage → part
Pricing	Rule engine (JSON price tables)	Itemised repair bill
Settlement	IRDAI depreciation + deductible rules	Insurer / customer share

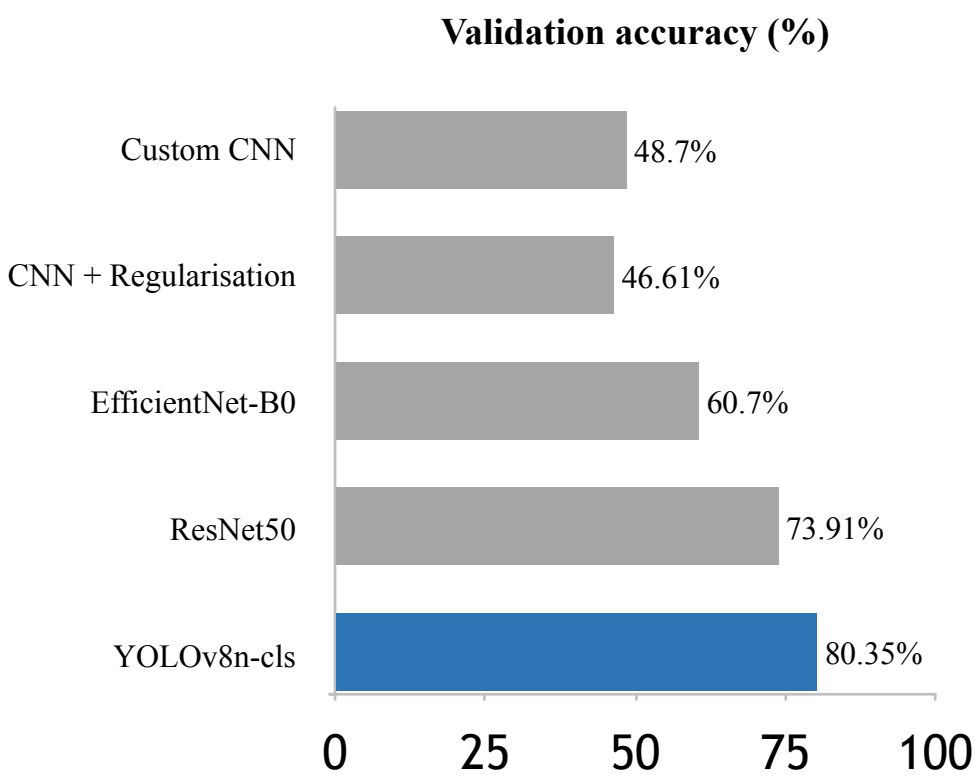
Compared for classification: custom CNN, CNN + regularisation, EfficientNet-B0, ResNet50 and YOLOv8n-cls, all on the same 2,300-image dataset.

9. Datasets



Dataset	Size	Used for
Car damage (front/rear × severity)	2,300 images, 6 classes	Classifier
Car parts & damages polygons (MIT)	998 + 814 images	Part & damage segmentation
Auto insurance claims (Kaggle)	1,000 claims, 40 fields	Analysis: $R^2 = -0.14$

10. Results: Classification



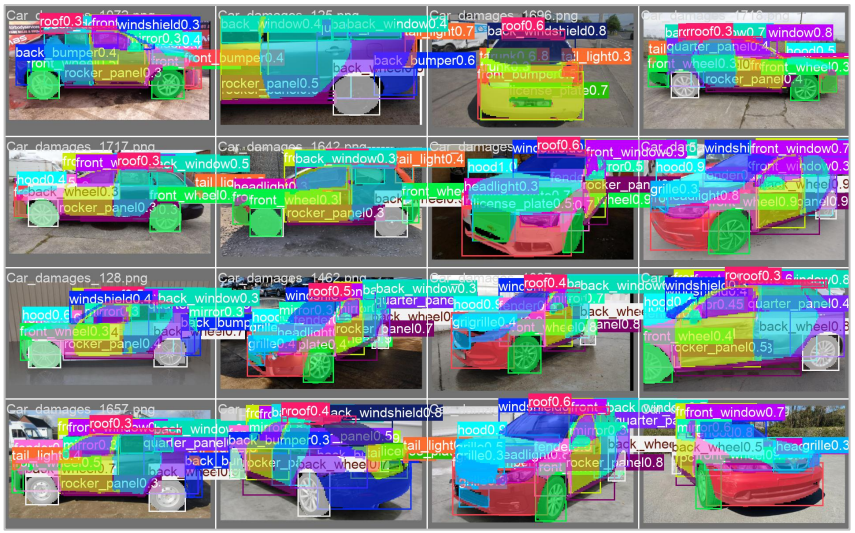
80.35%
YOLOv8n-cls accuracy

0.80
weighted F1-score

3.2 min
training on M4 GPU

- Best classes: front breakage F1 0.89, front normal 0.87; hardest: rear crushed 0.65.
- 575 held-out validation images; transfer learning from ImageNet.

11. Results: Segmentation

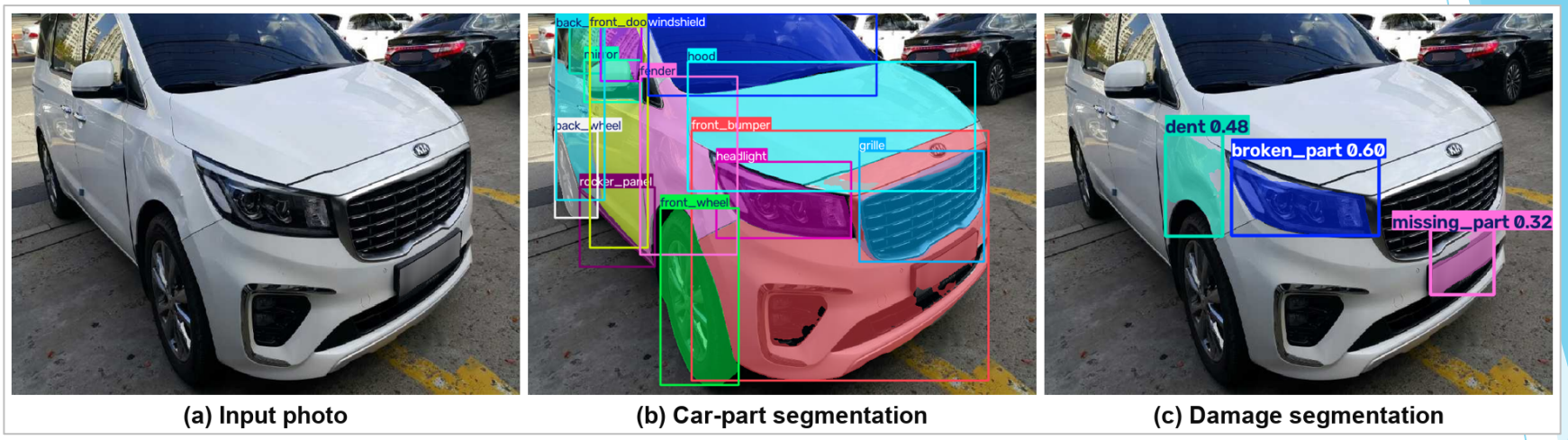


Car-part predictions on validation images

Metric	Parts	Damage
Classes	21	8
Mask mAP@50	**0.81**	0.15
Box mAP@50	0.83	–
Precision	0.81	0.41
Recall	0.75	0.18

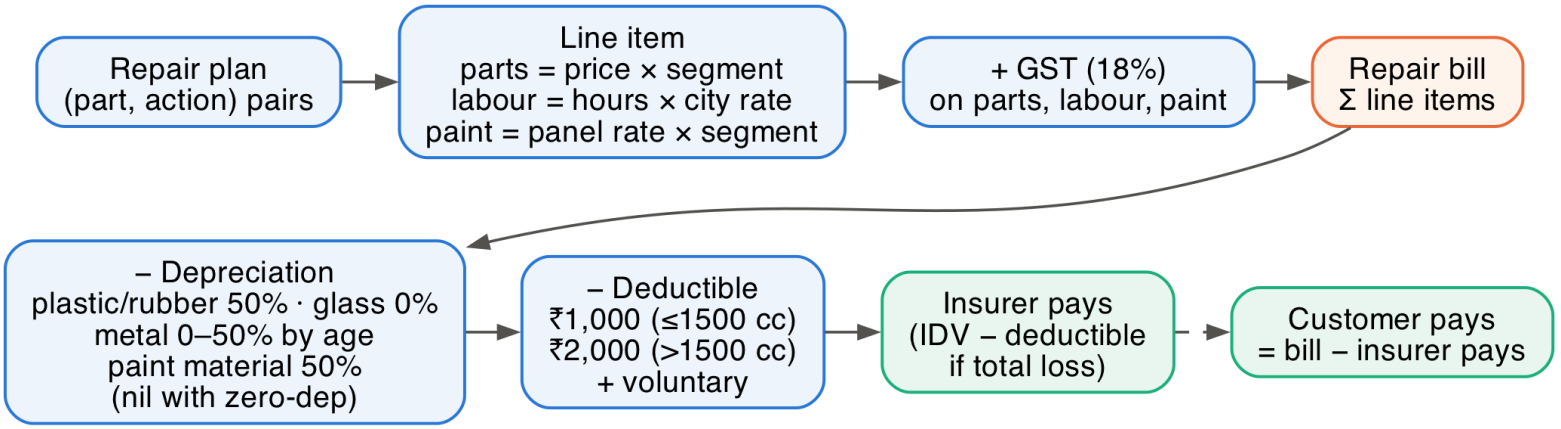
- Damage model is strongest on the **costly** types: crack 0.44, missing part 0.35, broken part 0.25.
- Thin damage (scratch, paint chip) is hard on 814 images; the severity classifier's high estimate covers what the damage model misses.
- End-to-end time: **~1.9 s per photo**.

12. Damage-to-Part Attribution & Claim



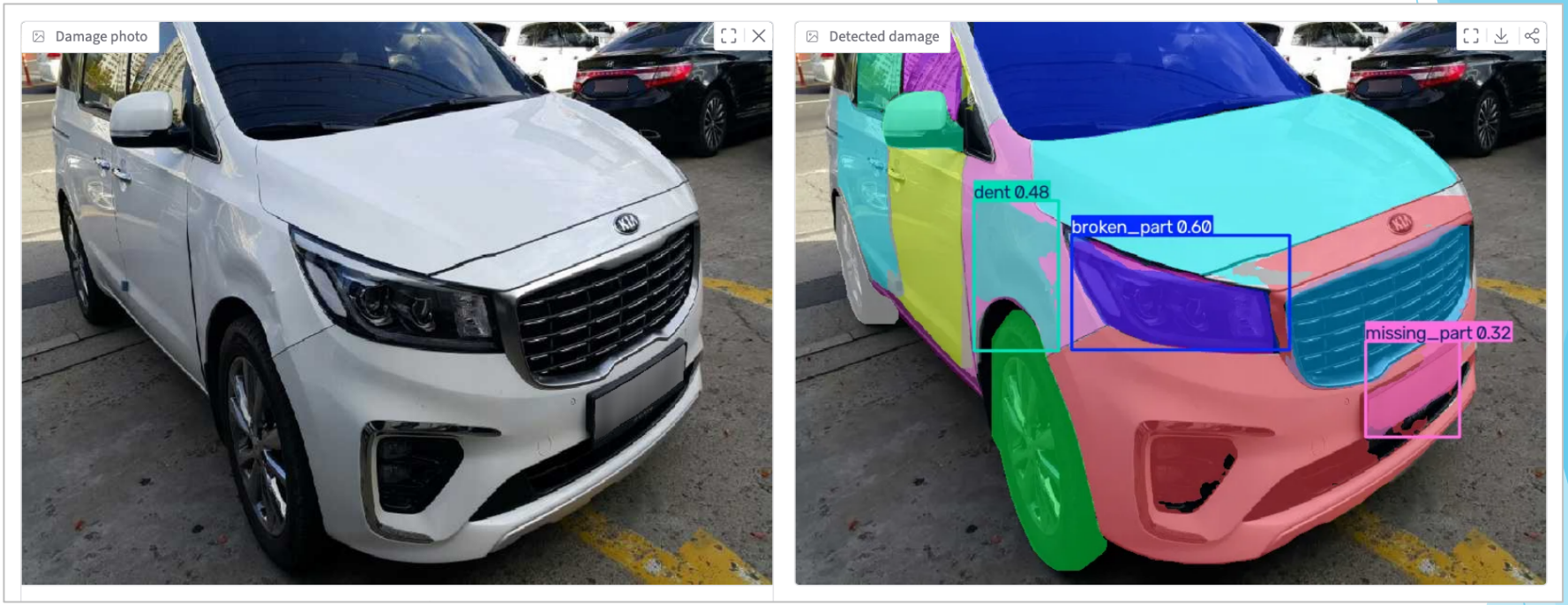
(a) Input photo (b) Car-part segmentation (c) Damage segmentation

Damage masks matched to part masks by overlap



Insurer pays = repair bill − depreciation − deductible (total loss if bill > 75% of IDV)

13. Implementation: Web App



- **Input:** photo, car segment, city tier, vehicle age, engine cc, IDV, zero-dep, voluntary deductible
- **Output:** annotated photo, damage table, low–high repair estimate, insurer and customer share
- **Stack:** Python, PyTorch (MPS), Ultralytics YOLO, Gradio; 32 automated tests passing

14. Worked Example

How SmartClaim AI Estimates a Claim (Step by Step)

1 Photo + Vehicle & Policy Details

User uploads one photo of the damage in the web app

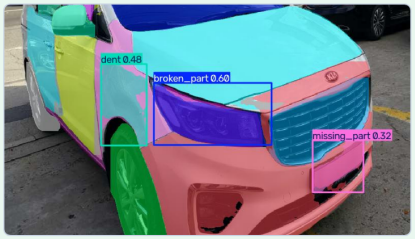


User enters details

Car segment	**Sedan**
City tier (labour rate)	**Tier 1**
Vehicle age	**3 years**
Engine capacity	**1,200 cc**
Insured Declared Value (IDV)	**₹6,00,000**
Zero-depreciation add-on	**No**
Voluntary deductible	**₹0**

2 AI Detection — 3 YOLO Models

Run on the Apple Silicon GPU (PyTorch MPS)



Model output (this photo)

① **YOLOv8n-cls** — location & severity
→ **Front · Crushed** 56%

② **YOLO11n-seg (21 car parts)**
→ headlamp, fender, front bumper, grille, bonnet, wheels…

③ **YOLO11n-seg (8 damage types)**
→ broken part 0.60 · dent 0.48 · missing part 0.32

3 Damage → Part → Repair Action

Each damage mask is assigned to the part it overlaps most (≥ 20%)

Damage	On part	Low plan	High plan
Broken part	Headlamp	Replace	Replace
Dent	Fender	Repair	Replace*
Missing part	Front bumper	Replace	Replace

* dent covers 43% of the fender (> 30%) → replaced in the high plan

High plan adds typical hidden damage for "Front · Crushed"
Grille · Bonnet · Radiator · Radiator support

Safety checks

- Front/rear look-alike parts corrected using the classifier's zone
- Scratches on glass / lamps treated as cosmetic (not billed)
- Confidence < 50% → "manual inspection" note

4 Pricing Engine + IRDAI Policy Rules

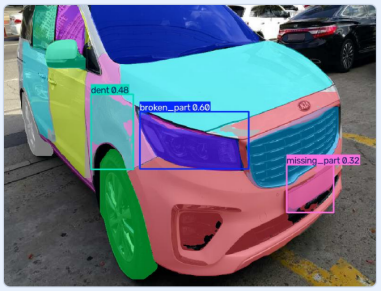
```
parts = base price × segment (sedan 1.35)
labour = hours × ₹650/h (tier 1)
paint = ₹2,800 × 1.35 per painted panel
+ GST 18% on parts, labour and paint
```

Low plan item	Parts	Labour	Paint	GST	Total (₹)
Headlamp (replace)	5,400	455	—	1,054	**6,909**
Fender (repair)	—	1,950	3,780	1,031	**6,761**
Front bumper (replace)	6,075	975	3,780	1,949	**12,779**

Apply policy rules (IRDAI / India Motor Tariff)

- **Depreciation** on replaced parts: plastic/rubber 50%, fibreglass 30%, glass 0%, metal by age (15% at 3 yrs); 50% on paint material — *nil with zero-dep*
- **Deductible**: ₹1,000 (≤ 1,500 cc) or ₹2,000 (> 1,500 cc) + voluntary
- **Total loss**: if bill > 75% of IDV → insurer pays IDV – deductible

5 Final Output in the Web App



Annotated photo with detected parts and damage · result in ≈ 1.9 s

Estimate and settlement (sedan, 3 yrs, 1,200 cc, IDV ₹6,00,000, tier 1)

	Low estimate (visible damage)	High estimate (+ typical hidden damage)
Parts in plan	Headlamp, fender, front bumper	+ grille, bonnet, radiator, radiator support
Repair bill (incl. GST)	**₹26,450**	**₹74,729**
– Depreciation (IRDAI)	₹7,885	₹15,333
– Compulsory deductible	₹1,000	₹1,000
Insurer pays	**₹17,564**	**₹58,397**
Customer pays	**₹8,885**	**₹16,333**
Total-loss check	₹74,729 < 75% of IDV (₹4,50,000) → normal repair claim	

Estimated Repair Range
₹26,450 – ₹74,729 (low – high)

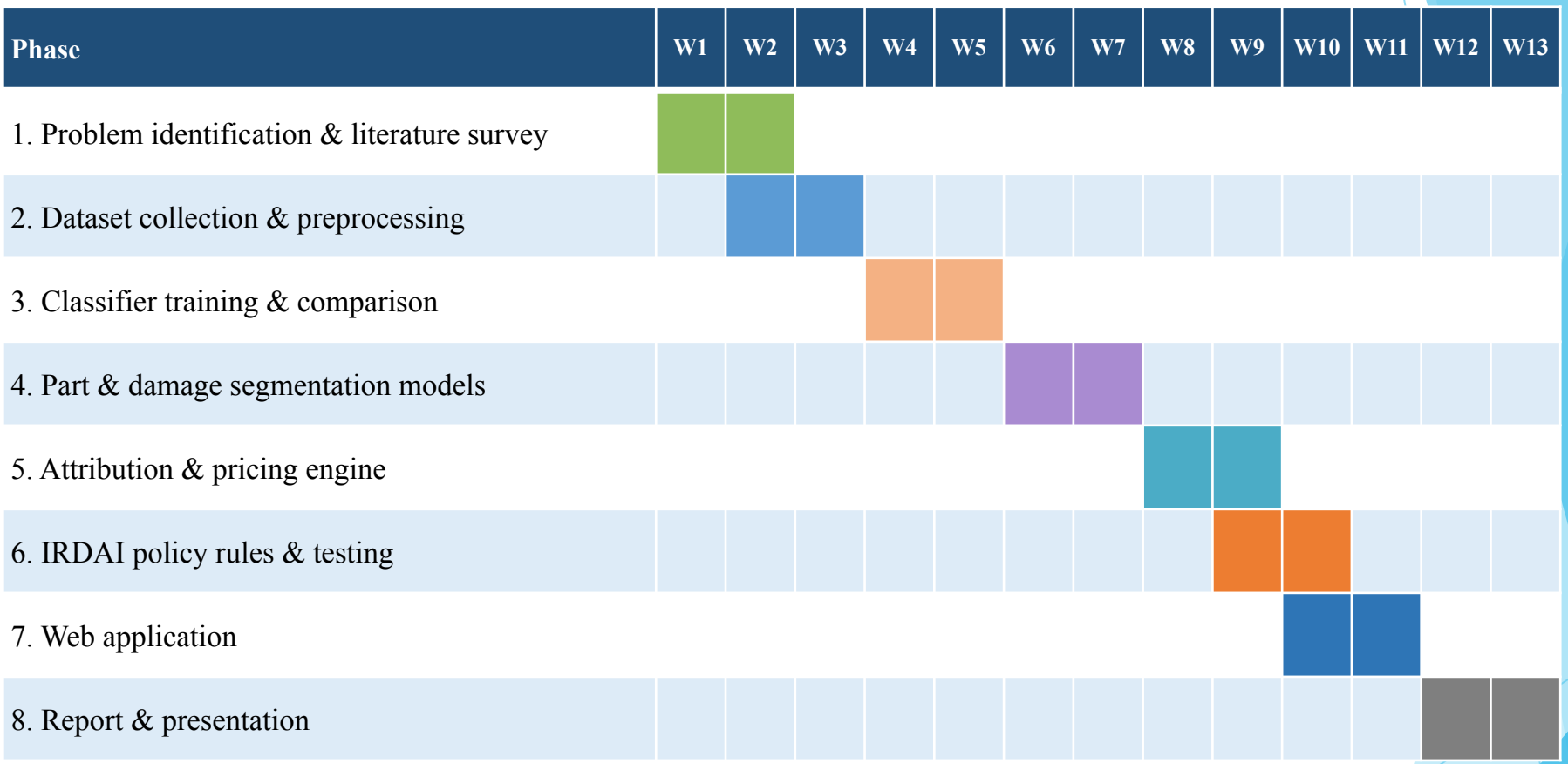
Insurer pays
₹17,564 – ₹58,397

Customer pays
₹8,885 – ₹16,333

Model confidence shown with the result
Classifier: Front · Crushed — 56% · Damage detections: 60% / 48% / 32%
Notes are added automatically for low confidence or when the classifier and damage model disagree.

1 Upload photo + vehicle & policy details → 2 Classify severity + segment parts & damage → 3 Attribute damage to parts → low / high repair plans → 4 Price items + apply IRDAI rules → 5 Annotated photo + estimate range + settlement

15. Project Timeline



16. Conclusion

- Built a complete pipeline from **one photo to a claim settlement** in about 2 seconds.
- Compared 5 classifiers; **YOLOv8n-cls** reached **80.35%** accuracy.
- Car-part segmentation with mask mAP@50 of **0.81** across 21 parts.
- Damage is attributed to **named parts**, so every estimate is itemised and explainable.
- Implemented **IRDAI** depreciation, deductibles, zero-dep and total-loss rules.
- Showed that historical claim amounts **cannot** price visible damage, which justified part-level pricing.

17. Future Scope

- Train on larger damage datasets for more epochs to improve scratch and dent detection.
- Combine **multiple photos** of the same car and cover side damage.
- Live part prices per make and model, and city-specific labour rates.
- **Fraud screening** of edited or re-used photos.
- **Mobile app** with guided photo capture and insurer integration.

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Thank You...!!